SMS MEER

SMS group

# WIRE ROD and BAR MILLS










# Moving forward for SATISFIED CUSTOMERS

## Tradition, leading know-how and innovations are united in the long company history of SMS Meer.

**1872** Founding of Maschinenfabrik Meer by the Meer brothers

**1926** Meer AG, 100% Mannesmann-röhren-Werke

**1955** Company is renamed Mannesmann Meer AG

**1975** Mannesmann takes over Demag AG

**1996** Takeover of the Tube Technology division of Innse

**1999** Merger between Mannesmann Demag AG Metallurgy and SMS Schloemann Siemag AG to form SMS Demag AG

**2000** Establishment of SMS Meer GmbH
Expansion of the product portfolio to include plants and machines for the aluminium industry (Hertwich Engineering GmbH)

**2005** Integration of the competence centre for continuous casters (Technica) into the Mönchengladbach operations

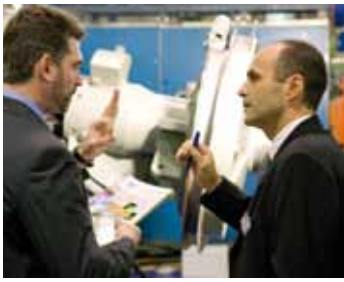
**2007** Expansion of the product portfolio to include closed-die forging, extrusion presses and ring rolling (SMS Eumuco)

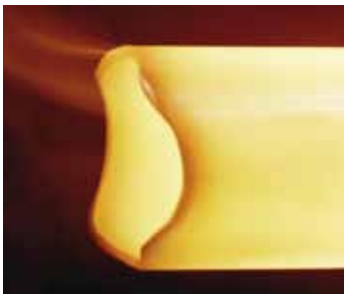
**2008** Takeover of the machine engineering activities of Schumag AG, expansion of the product portfolio to include drawing machines and bright steel production lines as well grinding and polishing machines
Expansion of the portfolio to include spiral pipe welding lines (PWS Automatisierungs- und Elektrotechnik GmbH)

## Company

SMS Meer is a division of the SMS group, world market leader in plant and machine engineering for the industrial processing of steel, aluminium and non-ferrous (NF) metals. Through organic growth and acquisitions of further leading companies, SMS Meer has developed into the leading full-line provider in the field of metal forming. With a product-spanning service organisation, the company offers its customers optimum support in all areas even after the commissioning of the machines and plants. With offices in Western and Eastern Europe, the Middle East and Asia as well as North and South America, SMS Meer cooperates with its customers around the globe. The annual turnover totals EUR 1.2 billion.







## Employees

SMS Meer has 2,400 employees, more than half of whom at the company headquarters in Mönchengladbach (North-Rhine Westphalia) – including 80 apprentices. As a family-owned company, SMS Meer practices the corporate culture of a medium-sized enterprise with flat hierarchies and short decision routes. The co-operation is marked by direct communication, respect and trust – while at the same time employing professional management methods and standardised processes.

## Products and customers

Whether tubes for pipelines, rails for the high-speed ICE and TGV trains, transmission components for the Mercedes S Class, the outer skin of the Ariane rocket or gigantic wind power generators: The machines and plants tailored by SMS Meer ensure the flawless production of the necessary parts of the highest possible standard. SMS Meer holds a roughly 40 percent share of the world market, making the company the undisputed world market leader for metal forming plants.















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# WIRE ROD MILLS
# for quality and stainless steels

The production of wire rod from quality and stainless steels is subject to constant change. The growing demands on the quality of the finished products and on the flexibility and cost-effectiveness of the plants necessitate new concepts and strategies from the plant manufacturer. This applies not only to the machine engineering, but also to new and innovative processes and technologies. The holistic consideration of the complete process is fundamental for meeting the customer's expectations.

SMS Meer satisfies these very high demands not only with new wire rod mills, but also through the revamping, modernisation and expansion of existing plants.

## SINGLE-STRAND HIGH-SPEED WIRE ROD MILLS

Single-strand high-speed wire rod mills from SMS Meer combine the rolling of small dimensions with high production rates – and that with low investment and operating costs. This guarantees a fast return on investment.

As a rule, such mills are designed for a production from 400,000 to over 800,000 tonnes per year. The typical product sizes lie in the range from 5.5 mm to 20 mm diameter, i.a. for low to high-carbon steels, cold-heading steels, free-cutting steels, alloyed steels, spring steels, ball bearing steels, welding wire, stainless steels and tool steels, in some cases with thermomechanical rolling. The optimum material properties of each finished product are achieved using facilities specially optimised or designed by SMS Meer.




Loop laying head



Coil store



Two single-strand high-speed wire rod mills




High-speed wire rod block

## THE TECHNICAL HIGHLIGHTS AT A GLANCE

- Product dimensions from 4.5 mm to 26 mm diameter
- Reliable rolling up to 120 m/s
- Tolerances of ± 0.1 mm and better
- Utilisation level over 90% when using FRS® (**F**lexible **R**eduction and **S**izing)
- Temperature-controlled rolling with the highest reproducibility for achieving the demanded material properties
- Thermomechanical rolling thanks to very high stand capacities, even with small roll diameters in all wire rod blocks
- Very good laying pattern even at high rolling speeds
- Individual drive of the wire rod block stands (MEERdrive®) permits very high flexibility of layout and roll sizing

# WIRE ROD MILLS
# for quality and stainless steels

## LAYOUT OF A WIRE ROD LINE FOR THERMOMECHANICAL ROLLING

A typical layout of a wire rod line for thermo-mechanical rolling looks essentially as follows:

- Heating furnace
- (High-speed) roughing train, with cantilever or housingless stands in alternating horizontal/vertical (H-V) arrangement.
- Free runout with holding conveyor
- Intermediate train, with cantilever or housingless stands in alternating horizontal/vertical (H-V) arrangement.
- Cooling and equalising sections (loop)
- 6-stand wire rod block with MEERdrive®
- 4-stand wire rod block with MEERdrive®
- High-speed loop laying head
- Loop cooling conveyor (LCC®) for low and high cooling rates
- Coil forming chamber with ring distribution system (RDS)
- Coil handling facilities with coil compactor



Cooling and equalising loop at Gerdau Acominas / Brazil




1 Heating furnace
2 High-speed roughing mill
3 Free run-out
4 Intermediate train
5 Cooling and equalising loop
6 6-stand wire rod block
7 4-stand wire rod block
8 Loop laying head
9 LCC®
10 Coil handling facilities

Typical layout



Loop laying head and loop cooling conveyor (LCC®)

## THERMOMECHANICAL ROLLING – WHY?

Thermomechanical rolling is the combination of a forming process with a heat treatment process. In particular, however, it is the rolling process during which the final forming lies between AC1 and AC3, so that no recrystallisation or only partial crystallisation of the austenite occurs.

The aim is to achieve a fine-grain finished microstructure with better mechanical properties. These properties can generally only be achieved – if at all – through extensive, time-consuming and energy-intensive post-treatment.

For example, the cold formability of cold heading grades can be improved by thermomechanical rolling, and at the same time a subsequent annealing and self-tempering can be partly or completely eliminated.

For ball bearing steels it is possible to reduce carbide separation by rolling at lower temperatures, resulting in greatly improved mechanical properties of the finished product.

For Cr, CrB and CrMo steels, the fine-grain microstructure produced by the TM process allows lower hardnesses to be achieved, eliminating the need for subsequent soft annealing.

Even with higher alloy steels and for steels without phase transformation, thermomechanical treatment can influence the microstructure so such an extent that further heat treatment is not necessary and significant improvements in properties can be achieved.

# BAR MILLS
# for quality and stainless steels

Growing demands on the tolerances and surface finish of the rolled stock require high standards in the design of the mechanical equipment. In order to guarantee top technical performance, quality and reliability, all the mill components involved in the production process are constantly analysed and optimised in close cooperation between all the engineering departments.

This form of cooperation has resulted in the world's first control system for bar mills the influence the finished product size online, and thus drastically reduces losses due to out-of-tolerance material – Automatic Size Control (ASC®).

Cooperation with the Tube Division of SMS Meer resulted in the first 3-roll block for bars with hydraulic roll adjustment (PSM® (Precision Sizing Mill)) that sets new standards with respect to tolerances.



▲ ASC® system in operation



Rake-type cooling bed

## THE TECHNICAL HIGHLIGHTS AT A GLANCE

- Product dimensions from 6 mm to 130 mm diameter
- „Free-size" rolling when using a PSM®
- Tolerances better than ¼ DIN when using a PSM® or an ASC® system
- Temperature-controlled and thermomechanical rolling with the highest reproducibility for achieving the demanded material properties
- Surface quenching and tempering from the rolling heat thanks to water cooling section downline of the finishing train
- High charge weights
- Cooling beds with high-speed cross-transfer unit for hot stacking



# BAR MILLS
# for quality and stainless steels

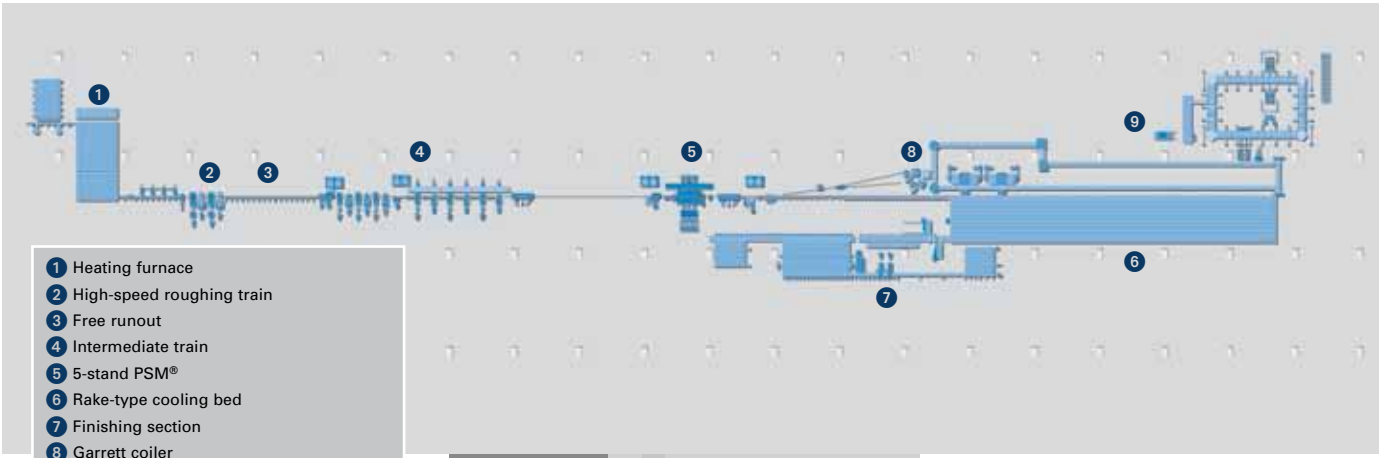
## LAYOUT OF A BAR MILL LINE FOR QUALITY AND STAINLESS STEELS

A typical layout of a bar mill line for quality and stainless steels looks essentially as follows:

- Heating furnace
- (High-speed) roughing train, with cantilever or housingless stands in alternating horizontal/vertical (H-V) arrangement
- Free runout with holding conveyor
- Intermediate train, with cantilever or housingless stands in alternating horizontal/vertical (H-V) arrangement.
- Cooling and equalising sections
- 5-stand PSM®
- Cooling section for „in-line" heat treatment
- Run-out roller table
- Rake-type cooling bed
- Finishing equipment
- Garrett coiler
- Coil handling facilities with coil compactor



Bar store




Typical layout




5-stand PSM® in operation



50 CrMo4, dia. 55 quenched and tempered from the rolling heat

## INLINE SURFACE QUENCHING AND SELF-TEMPERING – WHY?

Final rolling at temperatures of around 900°C followed by direct quenching produces a martensitic surface zone in steels which due to their chemical composition normally have a microstructure of ferrite and pearlite elements. This surface layer is then annealed again by the high temperature still prevailing at the core (in the temperature range between 600 and 700°C). This creates a microstructure that is ferritic-pearlitic at the core and has an annealed martensitic microstructure in the surface zone. This microstructure gives the steel its high strength characteristics, but at the same time good suitability for welding.

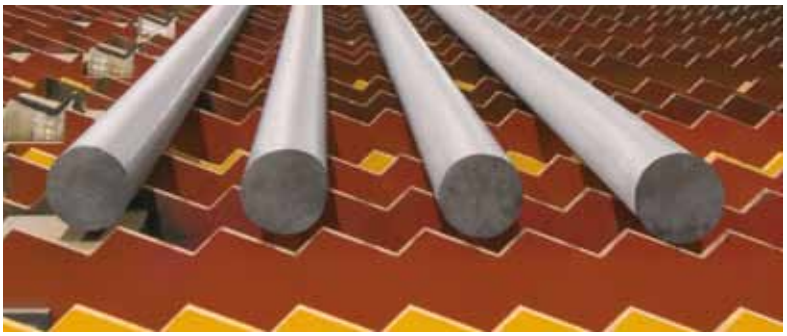
Further benefits of this process can be seen in that on the one hand, the costly addition of microalloying elements can be eliminated, and on the other hand the demands can be satisfied even with reduced carbon contents.

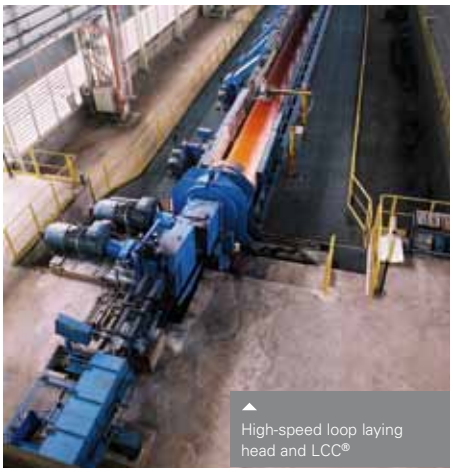
# COMBINED
# wire rod and bar mills

The components for wire rod and bar mills provide an exceptionally broad basis for combined mills that offer the owner very great flexibility with respect to the finished dimensions. No matter whether wire rod bundles, bars or wound coils – customers' wishes can be met quickly and flexibly.

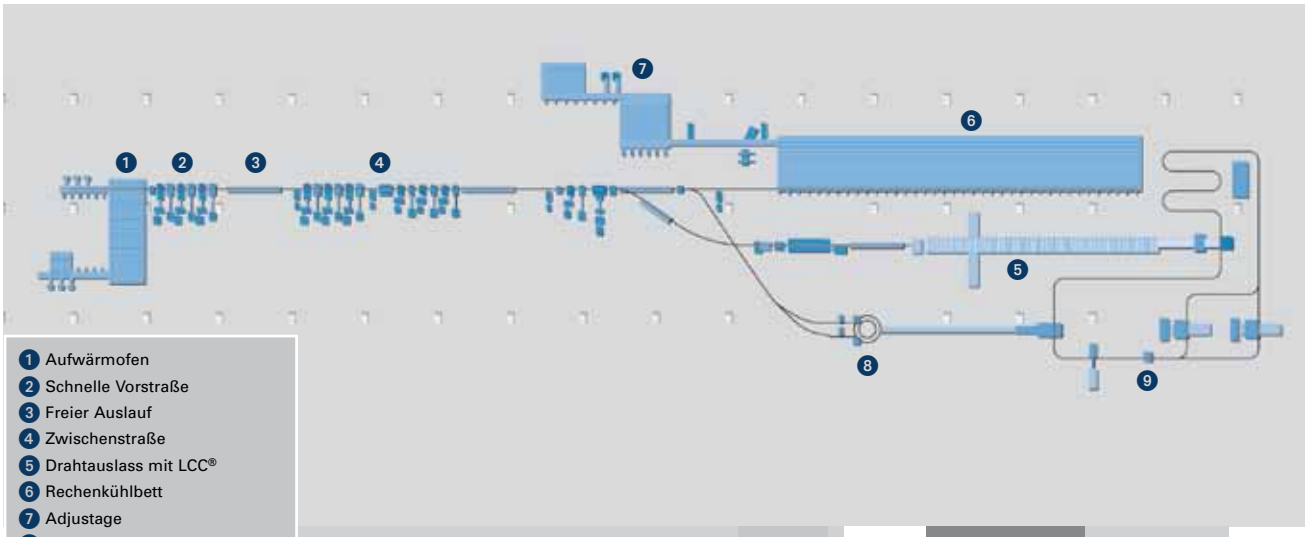
Whether for the planning of new combined wire rod and bar mills or for the modernisation of existing plants, attention has to be focussed on the following technical aspects:

- High production with a broad range of products
- First-class quality of the rolled products, not only with respect to tolerance, but also to surface finish and metallurgical properties.
- High flexibility of the line





▲ High-speed loop laying head and LCC®




▲ Typical layout



Combined wire rod and bar mill



Housingless roll stands



Garret coiler



Garret coil



Finished bar store



Coil store

# HOUSINGLESS roll stands

The housingless (HL) roll stand construction is the backbone of modern rolling mills. The modular design permits the use of HL stand cassettes in all possible configurations: horizontal, vertical, tiltable and universal.

The HL concept is thus suitable for roughing, intermediate and finishing trains. The stand sizes differ, depending on the necessary dimensions of the rolls and roll journals, pass schedule, pass form as well as the gearbox and motor characteristics. In order to determine all the process variables, the Engineering department at SMS Meer enters all the load data into special computation models and thus determines the optimum stand size.

The HL roll stands are employed in bar mills, wire rod mills, light section mills and combination mills.

The main features of the HL design are:

- Conservative criteria for the component design and general construction in order to achieve compactness and rigidity of the components.
- Low roll bending modulus (favourable ratio of roll journal diameter to roll diameter)
- Durable, multi-row roller bearings with self-aligning chocks under load
- Backlash-free balancing of the chocks
- Roller beams designed for simple and exact adjustment of the guides and guards
- The main operational benefits are thus:
- Finished product meets the demanded form and dimensional tolerances, hence close control of the metre weight
- Time saving for stand changes thanks to roll changing outside the rolling line
- Operational flexibility, use of the same stand unit in horizontal, vertical, tiltable and universal configuration, hence less spare part stocking
- Significant reduction in maintenance times and costs due to small number of components and easier accessibility
- Automated roll gap adjustment
- Integration into the fully automated control system



▲ Universal stand in operation

## MODERNISATION OF THE FINISHING TRAIN AT NUCOR DARLINGTON

Nucor Steel is the largest steel producer in the USA. The works in Darlington, South Carolina, is the Group's historical birthplace. The Group operates two bar mills. In 2008 SMS Meer received an order to modernise Mill No. 1. The new continuous intermediate/finishing train consists of four horizontal, one vertical and three tilting stands, all of housingless (HL) design. The tilting stands have a double drive with motors and gearboxes that remain in their position, irrespective of the stand configuration (H or V); this offers benefits for maintenance and operational safety of the plant, easier access to the installations with simultaneous reduction of the depth and size of the necessary foundations.



▲ Housingless roll stands in operation



## FLEXIBLE HIGH-PERFORMANCE WIRE ROD MILL AT GERDAU ACOMINAS, BRAZIL

Since 2003, Gerdau Acominas in Ouro Branco, Minas Gerais, Brazil, has been operating a high-performance single-strand wire rod mill for quality steels with a nominal annual capacity of 550,000 t. The main grades rolled are low, medium and high carbon steels, cold heading steels, rebar and low-alloy steels. SMS Meer supplied the whole rolling technology. Apatr from all the possiblities of temperature-controlled rolling, the mill is characterised in particular by its enormous flexibility. Compared with a conventional plant layout, this concept offers a potential increase in productivity of around 70,000 t/year. The delivered 160 mm square billets are rolled at up to 130 t/h, first in a 16-stand roughing and finishing train with a free runout from stand 4 onto a holding conveyor. These 16 HL stands are arranged as follows:

- 6 roughing stands, HL 630-26 in H-V arrangement
- 6 intermediate stands, HL 630-26 and 450-20 in H-V arrangement
- 4 finishing stands, HL 355-16 in H-V arrangement

The high-speed section then starts with 2 cantilever stands, Type 200, followed by an 8-stand wire rod block and a 4-stand FRS® (Flexible Reduction and Sizing) block. The whole size range from 5.5 up to 22 mm is finish-rolled in a „one-family" pass on the FRS®. Finishing block and FRS® are designed for minimum inlet temperatures of 750°C which, combined with the „LOOP" technology, provide optimum preconditions for thermomechanical rolling. A total of 14 water cooling boxes and the 105 m long loop cooling conveyor (LCC®) guarantee optimum temperature control of the products.

# CANTILEVER roll stands

Cantilever (CL) compact stands are employed in a wide range of sizes for a vast variety of applications. Preferred fields of application are:

- 1-strand mill lines in horizontal/vertical arrangement
- In split intermediate trains of two or more strand mill lines
- As replacement stand for mill train modifications (e.g. in the event of space problems)
- As prefinisher stands in wire rod delivery sections
- In finishing trains

## BENEFITS

The cantilever compact stands with roll shafts mounted in floating bearings have the following advantages:

- Low foundation costs
- Low rolling costs
- Low stocks of change and spare parts
- Short roll dressing times
- Short maintenance times
- Fast roll and stand changing
- Optimum accessibility
- Short erection time with low erection costs

## SPECIAL DESIGN FEATURES

- Stand types with roll diameters from 160 mm to 900 mm
- Compact design with low weight
- Ease of operation and maintenance
  No necessity for axial roll ring adjustment
- Rigid stand construction
- Large roll shaft diameter (roller bearings)
- Low roll bending
- Stand of cassette construction
- Cassettes of the same stand type are interchangeable, even between horizontal and vertical stands
- Safety coupling as overload protection
- High load-bearing strength even with small diameters, hence ideally suited also for use in high-speed wire rod blocks, even at low initial pass temperatures



Cantilever stands with changing equipment



Cantilever stands in an intermediate train




Cantilever stands in a roughing train

## COST BENEFITS:

Lower investment costs thanks to:

- Reduced space requirement for the mill
- Small and simpler foundations
- No guide troughs between roughing train stands necessary
- Savings with respec to roll ring machining facilities, as small types are sufficient
- Small roll ring stocks
- Less major spare parts (such as spindles)
- No change parts such as roll assemblies or complete stands
- Shorter erection times

Lower operating costs thanks to:

- Less spare parts
- Shorter changing times for roll ring and programme changes
- Less personnel in the stand workshop
- Higher material yield factor

# The LOOP technology

## COOLING AND EQUALISING LOOP

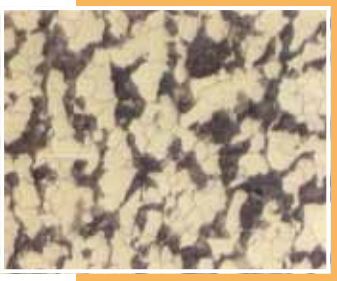
For final rolling at low temperatures in wire rod mills, the material entering the wire rod block must be intensively cooled. This must then be followed by a sufficiently long equalising section to allow the metallurgical properties to be achieved uniformly over the cross-section of the finished product. Without equalising section, the temperature difference between surface and core can be so large that different microstructures may be created during the subsequent forming process.

On the other hand, there are steel grades that have to be rolled as hot as possible and for which a long equalising section leads to a deterioration in quality. The loop technology allows these two demands to be perfectly reconciled.

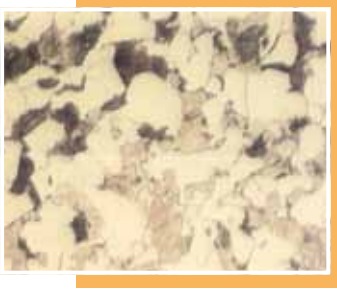
From the intermediate train, the material can take the short, direct route or the route via the loop with additional water boxes and long equalising sections before being rolled in the wire rod finishing block.

### ROLLING RESULTS:

The pictures show the effects of the loop on the microstructure of 16MnCr5 steel in 500x magnification.



This picture shows conventional rolling with a grain size of ASTM 8 and a composition of ferrite + pearlite + bainite + martensite.

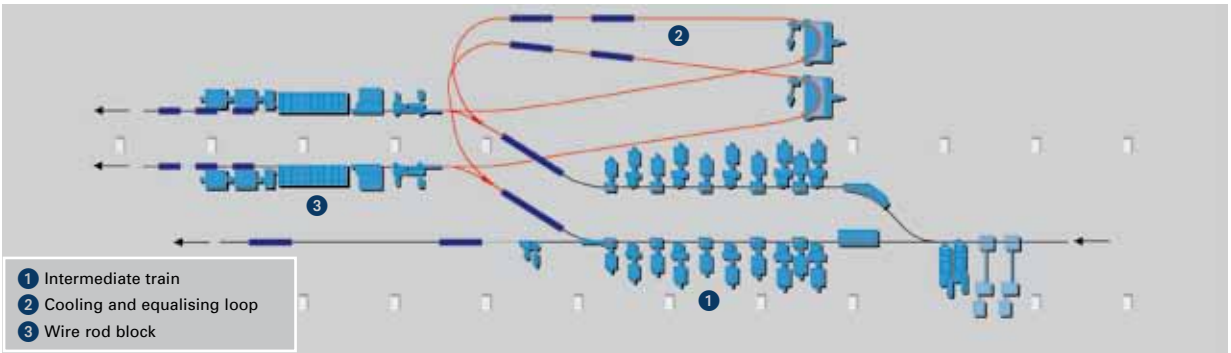


This picture shows thermomechanical rolling with a grain size of ASTM 9 – 10 and a composition of ferrite + pearlite (free from bainite + martensite).



▲ Cooling and equalising loop at Voest-Alpine

Typical layout ▼



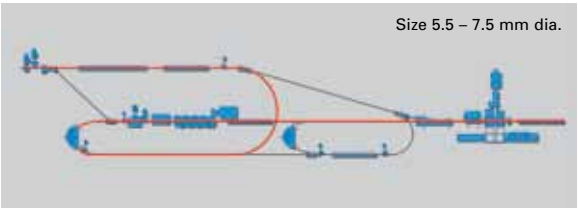


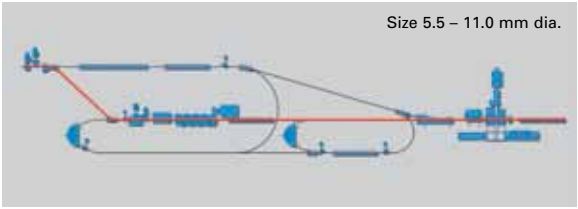
Multiline loop at
Gerdau Açominas/Brazil

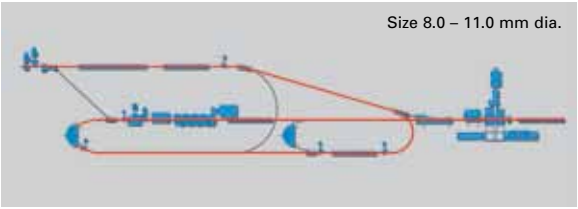
## MULTILINE LOOP

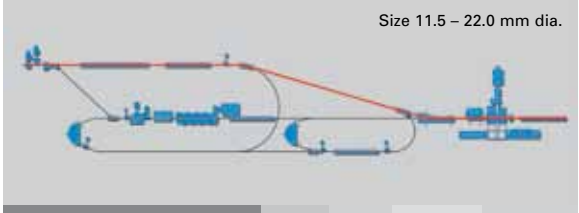
In wire rod mills in which all sizes are finish-rolled with one FRS®, a multiline loop is an optimum variant. When rolling finished sizes larger than 11.5 mm, the wire rod block upline of the FRS® is no longer used.

The multiline loop allows the wire rod block to be changed over for the next rolling while these sizes are still being rolled. The advantage of being able to roll hot or cold, however, is still retained.

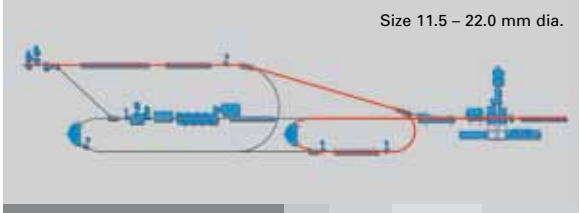



Use of the multiline loop
for normal rolling




Use of the multiline loop
for thermomechanical rolling

# WIRE ROD BLOCK and FRS®

In wire rod mills, the wire rod block represents one of the key elements. Only through its development has it become possible to safely roll thin wire rod formats at speeds of over 120 m/s.

The trend in wire rod rolling, however, is no longer just to even higher speeds but also to the rolling of ever higher grades with a simultaneous improvement in the metallurgical properties.

It was these demands that led to the development of the FRS® (**F**lexible **R**eduction and **S**izing) block. This is a four-stand block with speed-shift gearboxes that is installed downline of a wire rod block. On this FRS®, all dimensions between 5 and 26 mm diameter can be finish-rolled with the advantage of having created a real "one-family rolling". This means that only one pass size is used in each stand over the whole size range, and the necessary ingoing cross-sections to the FRS® are rolled by simply inserting or removing two passes. Thanks to the cooling section in between, this arrangement permits thermomechanical rolling of all sizes between 5 and 26 mm diameter with less work and time for roll changing.

The finishing block and FRS® have the following design features:

- Compact design
- Less vibration
- Lower noise level
- Guide design for „controlled cooling technology" and inter-stand cooling of round profiles
- Design for high speeds (up to 140 m/s) and high roll forces with UHD/HD (ultra-heavy duty/heavy-duty) stands for minimum initial pass temperatures of 750°C.
- A symmetrical roll gap adjustment is ensured by simultaneous turning of the eccentric for each roll pair.
- No axial adjustment of roll rings necessary, as the roll rings remain precisely positioned on the roll shafts during roll ring changing thanks to the use of a fixed oil splash ring
- No movement of the roll rings on the roll shafts thanks to the special construction for fixing the tapered bushes on the shafts
- Bar tracking from the inlet side of the wire rod block through to the loop laying head for fault detection (electrical equipment) as protection against faulty rollings
- „Safety cord" in the vicinity of the rolling line of the wire rod block signals the occurrence of a faulty rolling, the control circuit activates an emergency shear upline of the block.



10-stand wire rod block



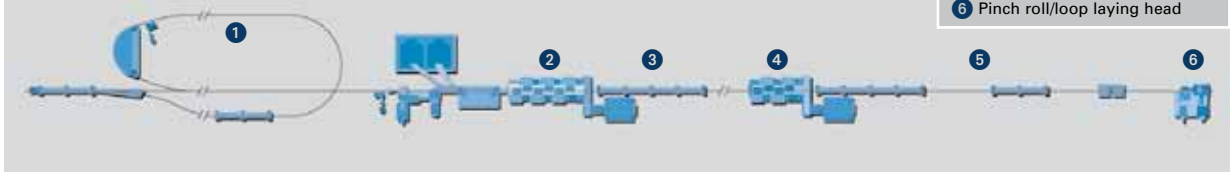
4-stand FRS®



4-stand wire rod block

Typical layout for the new wire rod mill concept

1 Cooling and equalising loop
2 6-stand wire rod block
3 Water cooling section
4 4-stand wire rod block
5 Water cooling section
6 Pinch roll/loop laying head

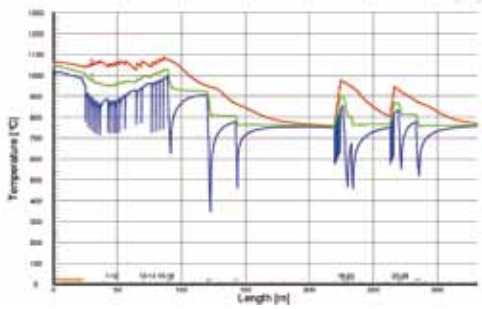



## NEW WIRE ROD MILL CONCEPT

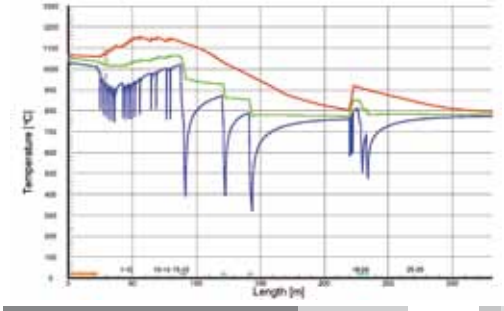
The use of a 6-stand and a 4-stand wire rod block allows all wire rod dimensions to be rolled thermomechanically and inexpensively with high production rates. By splitting the former wire rod block, it is now possible to finish-roll any product size in max. 4 passes. With sufficient cooling and good temperature equalisation over the cross-section, thermomechanical rolling at high production rates is thus possible. A good precondition for this is the SMS Meer loop technology that is already in successful operation in numerous wire rod mills.

It allows the ingoing temperature into the first wire rod block to be reduced to 750°C, and that with a temperature profile of less than 50°C.

This is a precondition for being able to reach 750°C again before the last 4 passes for sizes that still have to be rolled with 10 passes, hence making normalising or thermomechanical rolling possible even for small dimensions.




Temperature curve for 5.5 mm dia., V=105 m/s, cold heading grade




Temperature curve for 16 mm dia., V=26.5 m/s, cold heading grade

# MEERdrive®

The MEERdrive® is a new development from SMS Meer that represents a revolution in the field of wire rod blocks.

Today, all the stands of a wire rod block or an FRS® are driven jointly by one or more motors in tandem arrangement via complex gearing systems.

Thanks to the „electronic gearbox" developed by SMS Meer it is now possible to control the motors of a wire rod block or FRS® relative to one another so that the stands function like a wire rod block and, furthermore, can be far more precisely controlled.

## SUCCESSFUL COMMISSIONING:

In January 2009, Siderúrgica Norte Brasil S.A. (SINOBRAS) commissioned the first block with MEERdrive®, a 10-stand block in H/V arrangement with a guaranteed maximum speed of 80 m/s.

Even shortly after commissioning, 5.5 mm wire rods were being rolled at a rolling speed of over 80 m/s with a tolerance of ±0.1 mm.

The MEERdrive® represents a revolutionary technology for the wire rod block, as the fixed reduction ratios between the stands can be eliminated. This offers enormous flexibility for the roll sizing, as a wide range of different area reductions can be rolled in the same stand. The roll ring management is also greatly simplified, as the roll diameters no longer require a constant ratio relative to one another, and hence the superfluous machining down of roll rings within a rolling family is eliminated. Furthermore, the number of passes required can also be reduced.

The splitter gearbox common in wire rod blocks today is eliminated, allowing investment and maintenance costs to be saved.

This flexibility means that when using the MEERdrive® on a four-stand FRS®, only one speed-shift gearbox is required per stand for the speed range and no longer the additional gearboxes for the change in the reduction ratios between stands .

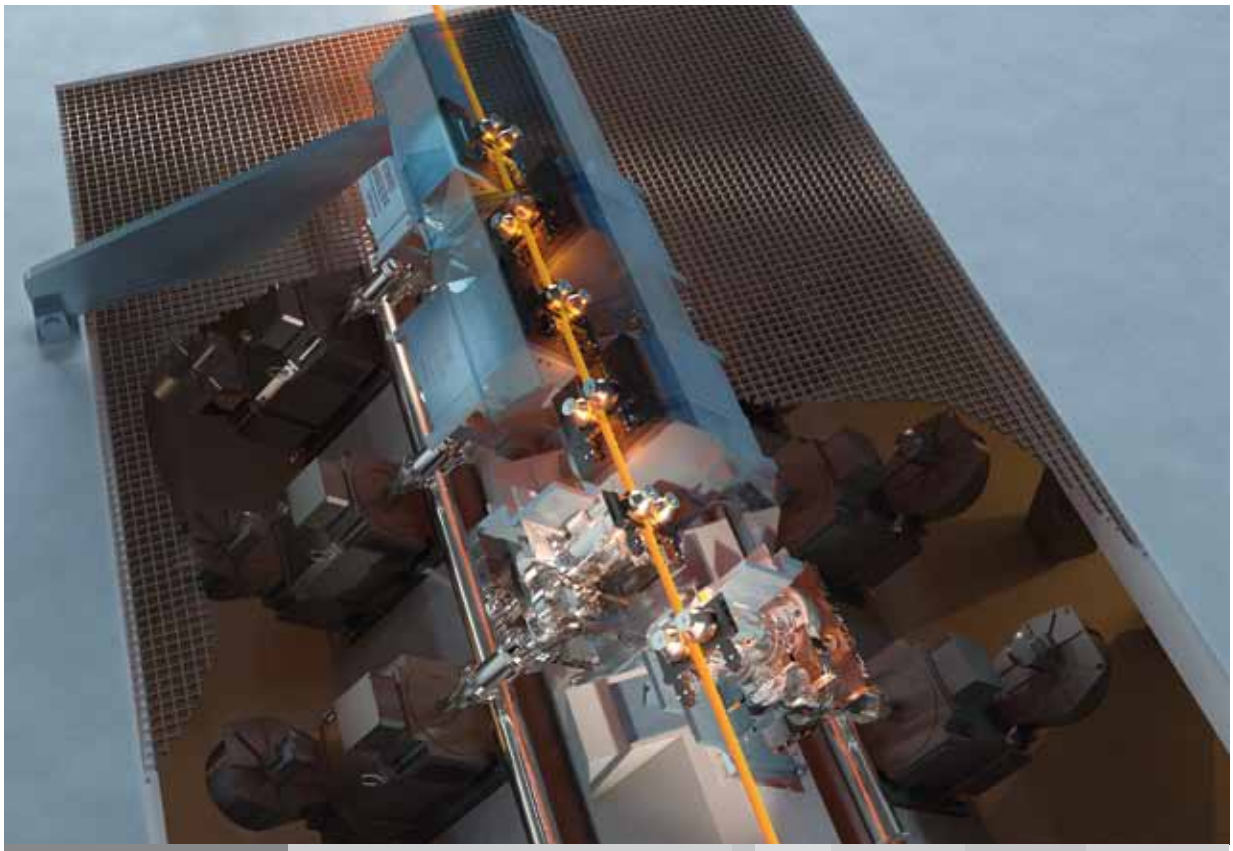
One-family rolling, thermomechanical rolling, finish-rolling of all sizes in the last four passes, two, four, six, eight or ten-stand blocks – all that presents no problem with the MEERdrive® and allows the owner to react far more flexibly and inexpensively to changing market demands. The MEERdrive® opens wide the door for the next generation of wire rod mills.

## ADVANTAGES:

- Complete flexibility in the sizing, as there are no fixed reduction ratios between the individual passes
- Simplification of roll ring management, as uniform machining of the rolling families is eliminated
- Reduction of the operating costs through better utilisation of the roll rings and reduction of the necessary roll rings, as well as through energy savings at unused roll stands.
- Cassette design with standardised stand size CL230, CL200 and CL160, depending on the requirements
- Flexibility in the arrangement as 2, 4, 6, 8 or 10-stand block or as FRS®



10-stand block with individual drives (MEERdrive®)



Model of a 6-stand wire rod block with MEERdrive®

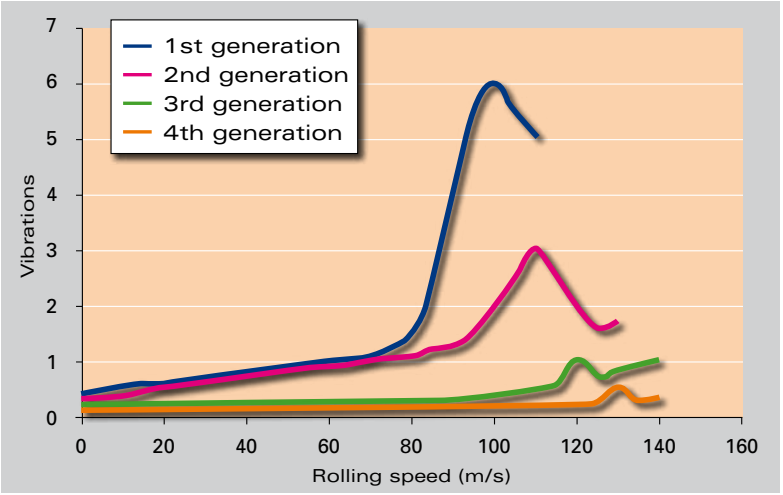
# LOOP LAYING HEAD

The demands on a loop laying head have changed greatly in recent years. The laying of different wire rod sizes in uniform loops with the loop laying head even at high rolling speeds remains an important criterion. In addition, however, the rolled grades and sizes have changed. A further difficulty is that the laying temperatures for certain grades have been greatly reduced for metallurgical reasons. In order to increase the profitability of the operated plant, the changing times for the wear parts, in particular the laying tubes, have to be reduced and the service life of the parts increased.

The latest 4th generation of the SMS Meer loop laying head meets all these demands. A quick-changing system and new, special laying tube materials improve the cost effectiveness of the system.

Thanks to a special bearing technology and improved bearings, vibrations have been drastically reduced (see chart).








Loop laying head



## TEST RUN

Every loop laying head has to undergo a test run lasting several days in our workshop before delivery. Vibrations and imbalances are measured in extensive test cycles. If the specified limits are not achieved, the loop laying head is modified at undergoes testing again.

Only loop laying heads meeting the high SMS Meer quality demands are delivered to our customers. This guarantees quick and successful commissioning of our machines on site.

# LCC® – loop cooling conveyors

The LCC® is one of the most important facilities in the whole process chain for achieving the desired material properties for a wide range of different steel grades. Apart from selected rolling temperatures, cooling settings and equalising sections, the optimum combination of speed, fan power and cover position on the LCC® has a crucial influence on the material properties. During „forced cooling", air is blown through the loosened windings with maximum fan power and open covers to cool the material as quickly as possible in order to achieve fine laminar pearlite.

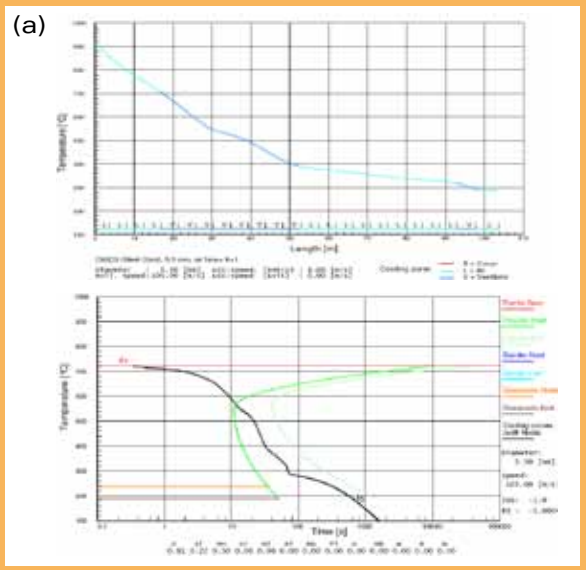
During „delayed cooling", the wire rod loops are transported without fan, with the covers closed and at low LCC® speed in order to keep the temperature in a given range for as long as possible. This enables a ferritic / pearlitic microstructure to be achieved.



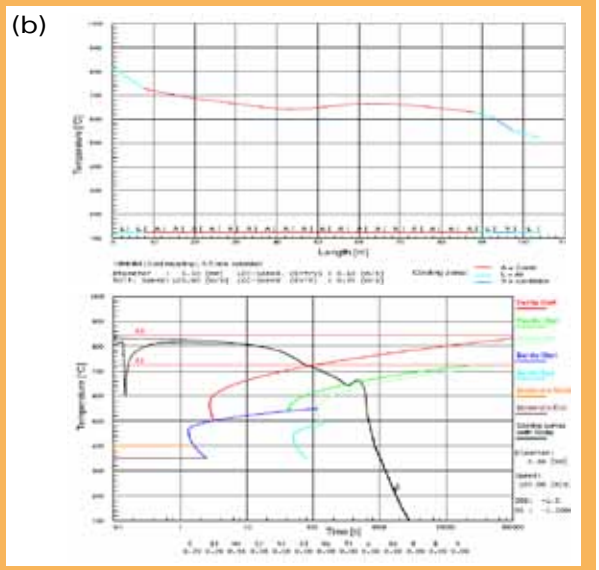
## EXAMPLES OF LCC® OPERATION

These figures show the LCC® operating modes „forced cooling" (a) and „delayed cooling" (b) with the corresponding TTT diagrams.

(a)



(b)

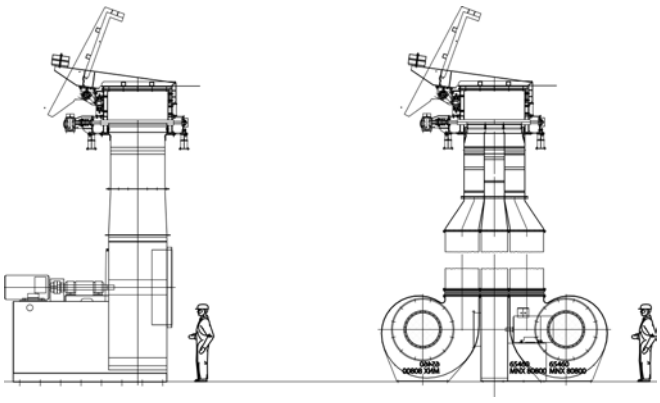


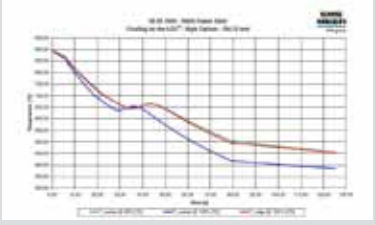


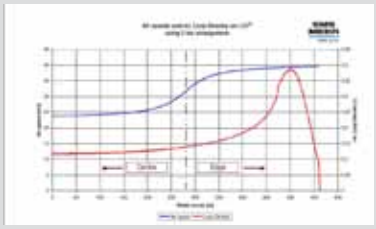
Loop laying head with LCC®

## NEW FAN TECHNOLOGY

Especially for the LCC®, SMS Meer has developed the „three fan technology". This technology guarantees a better temperature distribution over the individual loop windings. Through varying use of the fans, the cooling rate over the LCC® width can be individually adapted to the requirements.







# COIL HANDLING

SMS Meer coil handling is characterised by solutions adapted optimally to the customers' individual requirements. The modular design of the individual components permits high flexibility in the configuration.

## COIL TRANSPORT SYSTEMS

SMS Meer offers horizontal and vertical soil transport systems. As a special solution we also offer a "hybrid transport system" that utilises the respective benefits of the horizontal and vertical coil transport system for certain sections. A large proportion of identical modules is used for all these solutions.

## COIL COMPACTING PRESSES

SMS Meer also offers the matching coil compacting presses for the coil transport systems. Here again, a large number of parts of the horizontal and vertical coil compacting presses are identical. The modular configuration permits the use of wire or strap tying, depending on the customer's wishes.

## COIL TYING

SMS Meer offers wire and strap tying systems for all types of press. Here again, customers' wishes can be easily satisfied thank to the modular configuration of the systems. Different wire diameters can be used without changing over the press.




Vertical coil compacting press in our workshop



Wire tying




Horizontal coil compacting press in operation

## SPECIAL FEATURES:

- Universal, modular wire tying system for the use of tying wire with diameters from 6.5 to 8.0 mm.
- Universal, modular strapping system for the use of steel straps with widths from 25 to 32 mm.
- Extended protective devices to avoid scratching.
- Variable lifting tables capable of moving and positioning coils with outside diameters from 1100 to 1450 mm.
- Guaranteed nominal cycle time of 30 seconds



## LOW OVERALL DEPTH

The vertical coil compacting press is characterised by an extremely low installation depth. Foundation work and costs are significantly reduced and simplify the choice of installation location.

This also permits simple retrofitting in existing coil handling system with all types of pallet.

# PSM® – Precision Sizing Mill

The cooperation with the Tube Division of SMS Meer that has set new standards worldwide in the production of seamless tubes with the PQF® technology led to the development of the first 3-roll precision rolling block for bar steel with hydraulic roll adjustment (PSM®) that is now setting new hallmarks.

## KEY FEATURES:

- 3-roll block with 4 or 5 stand cassettes
- Modular design with standard stand cassettes in the sizes 350/380/435/480
- Hydraulic gauge control with the hydraulic capsule control system (HCCS) and monitor control
- Automatic adaptation to different rolling conditions
- „Free-size" rolling
- High roll forces (1250 kN)
- Max. finished product size: 130 mm diameter
- Quick inline changing of the stand cassettes (5 minutes)
- 3-roller guides with gas pressure dampers

## AUTOMATIC GAUGE AND MONITOR CONTROL

The PSM® has 4 to 5 cassettes each with three rolls which can be adjusted hydraulically even under load. This also permits the automatic gauge control (AGC) and monitor control in combination with a laser measuring system installed downline of the PSM® for dimension measurement. A CARTA® Process Model is use to set the process parameters.

## ADVANTAGES:

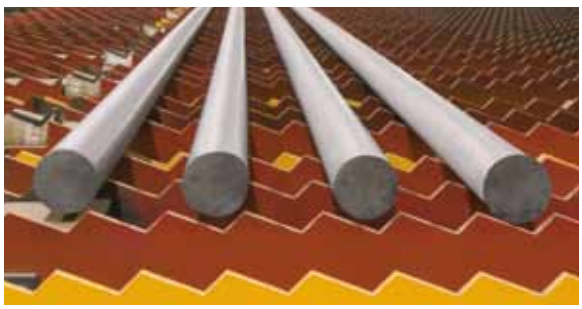
- **Precision rolling**
  Tolerance (≤ ± 0.1 mm) for all dimensions, steel grades and temperatures
- **„Free-size" rolling**
  Tolerance better than ¼ DIN for all dimensions, steel grades and temperatures
- **First bar within the given tolerance over the whole bar length**
  thanks to hydraulic gauge control with HCCS during rolling
- **Automatic compensation**
  of the stand spring-back and the initial pass variation
- **CARTA® offline process model**
  for automatic and optimised presettings
- **Thermomechanical rolling**
  possible thanks to the mechanical and electric design for a minimum initial pass temperature of 800°C up to a finished product size of 54 mm diameter
- **3-roller gas pressure guide**
  no adjustment necessary in „Free-size" rolling range
- **Machine protection**
  thanks to automatic opening of the rolls in the event of overload



3D section through a PSM® cassette




5-stand PSM® in operation at Timken/USA



## ONE-FAMILY AND FREE-SIZE ROLLING

Through the use of different finish-rolling cassettes in the PSM®, uniform sizing is possible in the roughing and intermediate train (one-family rolling), and infinitely variable production of all finished sizes (free-size rolling) is enabled by the gauge adjustment.

## SUCCESSFUL COMMISSIONING:

In September 2008, The Timken Company in Canton, Ohio/USA together with SMS Meer commissioned new equipment for the extension of the existing Harrison Steel bar mill. With this expansion, Timken has increased the mill capacity, extended the size range of its products and improved the product quality for the production of SBQ round bars in the range from 25 to 127 mm diameter. Tolerances of ½ ASTM (corresponding roughly to ¼ DIN) were achieved soon after commissioning. This reflects the high performance capability of the mill and documents the precision of the hydraulic roll adjustment systems.

# COILER SYSTEMS

Bar sizes up to 60 mm diameter can be produced in coils. Such coiler lines consist of a coil feeding line, two Garrett coilers, a coil transport facility with integrated coil handling facilities such as trimming and sampling station, and a compacting and tying station with weighing and labelling facilities.

The coiler operates with a tumbling motion and ensures that the coils produced achieve a high degree of filling stability and hence a stable form.

For thin dimensions and/or high coiling speeds, the installation of a spiral laying pot is often a benefit. During the coiling process, the spiral laying pot moves upwards together with the growing coil in order to control the distance between the coil feeder tube and the coil height. The result is high operational reliability together with a perfect form of the coil.

The design and construction of all these facilities ensures scratch-free surfaces of the top-class finished products.

## INLINE TREATMENT

For certain steel grades, a direct post-treatment from the rolling heat is expedient. On the one hand, this can be delayed cooling, on the other hand a forced cooling.

The delayed cooling is ensured by a hood-type cooling system in which the hoods of heat-insulating material are placed over the coil immediately behind the coiler. This process is suitable in particular for roller bearing steels, spring steels, steels for quenching and tempering and cold heading steels.

Forced cooling is achieved by the use of fans underneath the coil conveyor. At special blowing stations, air is blown from below through the coils at high velocity. This process is employed for low-carbon steel grades.

For the quenching and tempering of the coils, SMS Meer has developed a special coiler that can be filled with water, if necessary, and hence quenches the material. This process is particularly suitable for austenitic stainless steel grades.



▲ Transfer facilities to the coil conveyor



Coil arrangement



Special coiler in „dry“ operation



Special coiler in „wet“ operation

# PROCESS CONTROL SYSTEMS

SMS Meer has numerous references for the supply of the complete electrical and automation equipment for whole production lines. If the supplier of the mechanical equipment is also involved in the process control, this has obvious advantages. The close link between the design engineer on the mechanical side and his colleague for the automation system ensures a smooth process without communication problems.

The electrical and control modules from SMS Meer are characterised by the following features:

- Integrated and seamless process sequence between mechanical equipment and automation system
- Exclusive focus on and comprehensive competence in the design and supply of rolling mills and processing lines
- Choice of well-known components and systems, perfectly matched to the mechanical design
- Complete range of services, starting from the design and procurement through to the commissioning and personnel training
- Breakdown of services between Level 1 and Level 2, and integration into Level 3
- Easily integrated into existing automation systems
- Easy availability of commercial components
- Process models

## AUTOMATION FOR ADVANCED PROCESS APPLICATIONS

SMS Meer supplies control systems for high-tech applications, such as the PSM®, MEERdrive®, high-speed delivery system HSD®, the vertical coil compactor VCC® and controlled cooling technology (CCT®). In these systems it is crucial to combine the design of the mechanical and automation equipment in such a way that a high performance level can be achieved. Very short cycle times demand both outstanding mechanical components and advanced and high-speed automatic control systems. The software developed by SMS Meer perfectly matches the mechanical systems so that the performance demands are satisfied.

SMS Meer is thus ideally equipped to integrate new systems seamless into existing plants.



Main control desk of a VCC® line




Main control desk of the PSM®

## INTEGRATION OF MECHANICAL AND AUTOMATION SYSTEMS INTO EXISTING PLANTS

Most projects worldwide focus on the modernisation of existing production facilities. One example:

- Die Timken Company, Canton, Ohio/USA placed an order with SMS Meer in 2007 for the supply of a 3-roll precision sizing mill (PSM®) together with a new cooling bed and finishing equipment, including grinding-type cut-off saws. SMS Meer supplied the control system for the complete scope of supply and the integration into the existing control system.




Screen display

# PROCESS MODELS

## TEMPERATURE CONTROL

One example of an open-loop and closed-loop control system from SMS Meer is the controlled cooling technology (CCT®) already successfully installed numerous times. With this system, the whole temperature curve in a wire rod or bar mill from the furnace through to the finished product can be monitored, controlled and reproduced. It consists of the packages CCT Offline and CCT Online.

CCT Offline is a planning program for the creation of cooling programs and calculation of the settings and plant parameters fort he water cooling sections and the LCC® that can be directly integrated into the control of the line. CCT Offline also incorporates simulation cores for the calculation of the whole temperature curve, including LCC®, as well as a microstructure models that allows i.a. TTT diagrams and flow curves for the specific rolling lots to be generated.

CCT Online can be installed as an add-on. It is a process computer system for control and regulation of the cooling sections in the rolling line, including the LCC®. The system ensures compliance with the target temperatures of the cooling programs created in CCT Offline. The control system operates inline, thereby also minimising temperature fluctuations over the product length. Furthermore, irregularities from the heating process can be reduced and the coil quality becomes more homogeneous.

In addition, CCT Online offers extensive Level 2 functions, i.a. material tracking, report generation, data recording and logging, visualisation, graphic display of important measured values. Furthermore, it calculates the minimum possible pause time for setting the LCC® parameters for the cooling program change. This gap time minimisation (GTM) eliminates the otherwise necessary emptying of the LCC® and hence increases productivity.

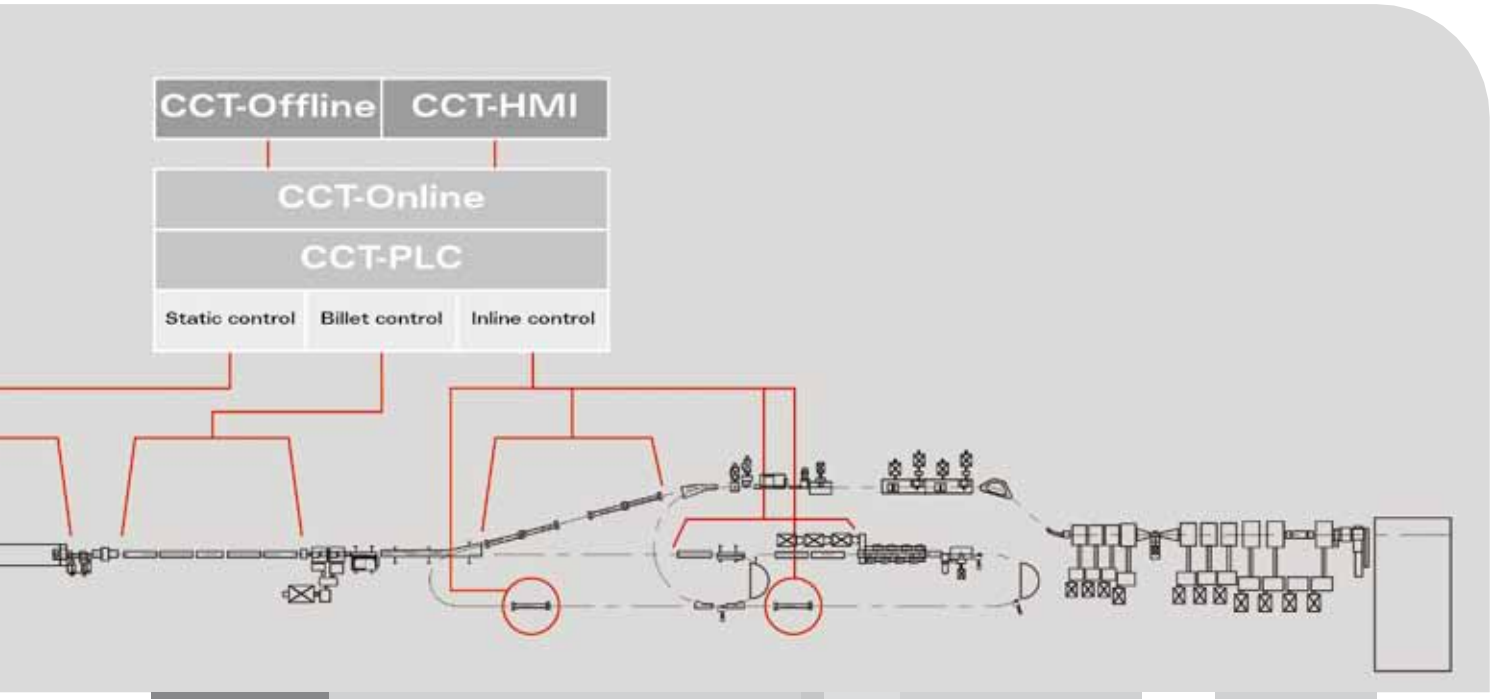




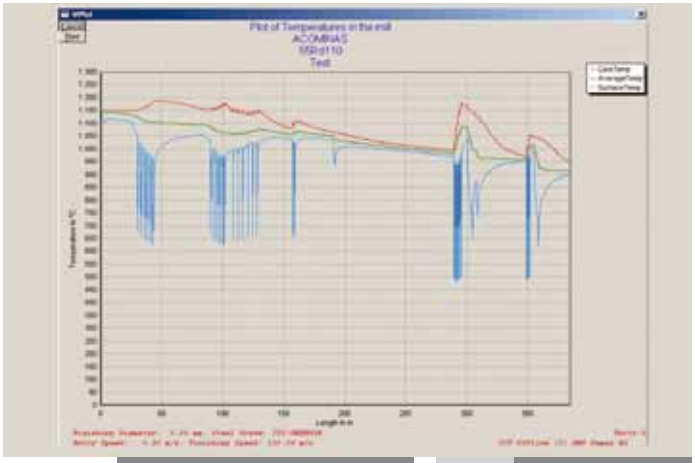

CCT-Offline: Microstructure model – flow curve range




CCT-Offline: Microstructure model – TTT diagram

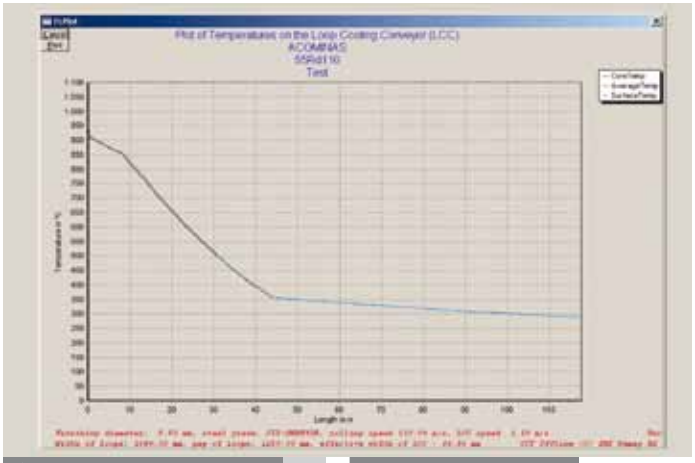



CCT® configuration



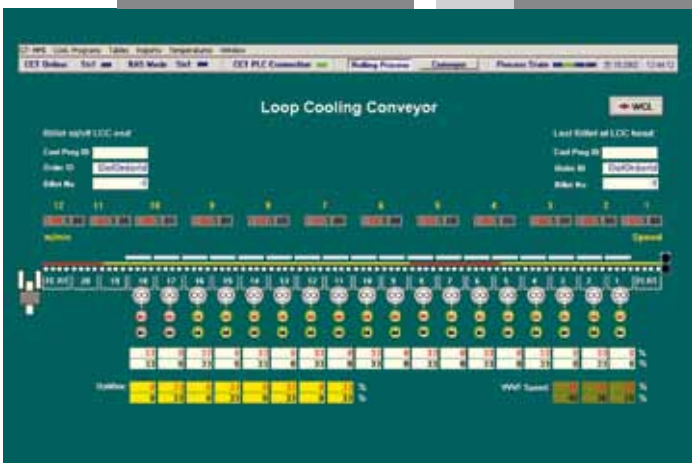
CCT-Offline:
Temperature curve in the line

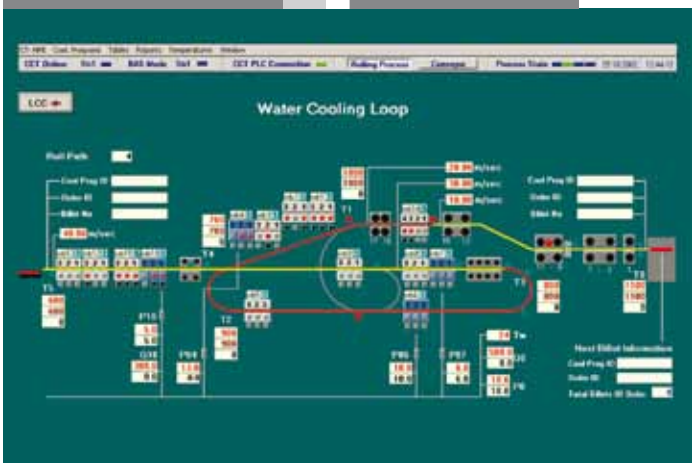
CCT-Online:
LCC® visualisation



CCT-Offline:
Temperature curve on the LCC®

CCT-Online:
Wire rod mill visualisation



# SMS MEER SERVICE – customer support a „plant" life long

SMS Meer develops and installs plants and machines. But not only that: The central Service Product Unit aims to ensure long-term customer satisfaction and therefore offers comprehensive support for plants and machines over their whole life cycle.

The staff know every detail and every single component. After all, they developed and designed it themselves. That is why the experts from SMS Meer are exactly the right choice when it comes to ensuring lasting reliable operation of the plant. But there's more: The support from the SMS Meer Service experts ensures that the plants are regularly modified to meet changing market demands.

The advantages for the customer of the service provided by SMS Meer are obvious:

- Increased productivity
- Increased plant availability
- Improved product quality
- Reduced operating costs
- Safeguarded plant value



Machines and plants from SMS Meer are in operation worldwide, and the organisational structure of Service is oriented accordingly. International teams and worldwide service centres guarantee immediate proximity to the customer, great flexibility, short response times and bundled competence for all-round customer support.

## INSTALLATION AND COMMISSIONING

A new or modernised plant can only play out all the advantages for the owner and ensure the expected added value if it operates flawlessly from the very beginning. The Service Product Unit at SMS Meer takes over everything that is necessary here: From the site management and OEM commissioning, through the installation of the mechanical, hydraulic and electronic equipment right up to the necessary performance tests. As an additional service, SMS Meer also offers the training of the customer's operating and maintenance personnel.

## START-UP SUPPORT

The SMS Meer experts on site guarantee their customers a stable production process from the very start. The SMS Meer experts and their customers can thus quickly remedy any start-up problems that could lead to costly faults or standstills. Trouble-free production and optimum product quality can thus be assured.

## OEM SPARE PART SERVICE

The high-quality OEM spare parts from SMS Meer are precision manufactured individually for all the machines and thus make a major contribution to the long-term safeguarding of the plant value. Thanks to the close links between Service and Research & Development (R&D) at SMS Meer, improvements and innovations are continuously integrated into the further development of the spare parts.





## EQUIPMENT CHECKS

With modern equipment and precise measuring methods, the SMS Meer experts track down improvement potentials in the customers' plants. Regular inspections also help to prevent an unscheduled standstill of the plants and machines.

## MODERNISATION

SMS Meer plants and machines are designed practically „for eternity". But: Thanks to the great innovativeness of the company, the technical possibilities never come to an end. The Service Product Unit ensures that even machines and plants that have already been in operation for many years can be brought to the state-of-the-art. That improves the performance of the plant – and at the same time increases efficiency by reducing operating costs.

## MAINTENANCE SERVICE

On request, the SMS Meer service experts draw up an individual maintenance concept for each customer. An extremely attractive offer – compared with the costs that can result from a standstill of the plant. The maintenance services range from comprehensive advice through individual maintenance contracts rights up to the takeover of the complete maintenance as part of a full-service.

## TRAINING PROGRAMMES

Standstill is a step backwards: Just as the machines and plants from SMS Meer are continuously developed further, so the know-how of the plant operators should be constantly kept up-to-date. Specially developed training programmes ensure the necessary know-how transfer.

## SMS MEER SERVICE – HELPING TO PRODUCE RELIABLY AND EFFICIENTLY

**MEETING** your **EXPECTATIONS**

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